INCH-POUND

A-A-50113A
February 21, 1990
SUPERSEDING
A-A-50113
March 31, 1987

COMMERCIAL ITEM DESCRIPTION

PITON, MOUNTAIN, ICE

The General Services Administration has authorized the use of this commercial item description in preference to type IV of MIL-P-1474.

1. CLASSIFICATION

1.1 Classification. This commercial item description covers ice pitons of the following types:

Type I - 6-5/8 ± 3/8 inches
Type II - 8-5/8 ± 3/8 inches

2. SALIENT CHARACTERISTICS

2.1 Design and construction. The pitons shall be chromium-molybdenum alloy 4130 steel. The eye shall be of sufficient size to accept a maximum 5/8 inch diameter snaplink and be permanently and firmly affixed to the top of the piton. The piton shaft shall be tubular in construction. The tip shall have milled or hand ground teeth to create sharp points so it will grab when screwing the piton into ice. The pitons shall have right hand threads in order that they will penetrate when turned clockwise into the ice, and shall be capable of withstanding a pull of 565 pounds when tested as specified in 3.4. The color shall be either natural or silver, and the manufacturer's identification symbol shall be permanently marked on the piton. Pitons shall be similar to figure 1, which is for illustrative purposes only and is not intended to preclude procurement of pitons otherwise meeting all the requirements of this document.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014.

AMSC N/A FSC 8465

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2.2 Type I. Type I pitons shall be 6-5/8 ± 3/8 inches in overall length. Maximum weight shall be 4.41 ounces.

2.3 Type II. Type II pitons shall be 8-5/8 ± 3/8 inches in overall length. Maximum weight shall be 5.12 ounces.

3. QUALITY ASSURANCE

3.1 Certification. The contractor shall certify, and maintain substantiating evidence, that the product offered meets the salient characteristics and requirements of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

3.2 Visual examination. Each lot of pitons shall be inspected in accordance with MIL-STD-105. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0. The lot size shall be expressed in units of pitons. The sample unit shall be one piton. The pitons shall be examined for the defects listed below:

Finish not smooth and adherent; not free of burrs, rough spots, sharp edges, slivers, fillets not rounded smoohtly; not free of forging lap; blade tapers not uniform; flat areas or projections; any component missing or not specified type; any component fractured, split, dented, bowed, malformed, damaged or loose; rivets loose and not properly peened; threads not uniform; markings in wrong location, incomplete, illegible, incomplete, or not as specified.

3.3 Dimensional examination. The pitons shall be examined without damaging or disassembling the pitons. Any dimension not within the specified tolerances shall be classified as a defect. The lot size shall be expressed in units of pitons. The sample unit shall be one piton. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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3.4 Testing. Both type I and II ice pitons shall be able to withstand a minimum pull of 565 pounds tangent to the major (long) axis of the ice piton. The force shall be applied at the eye while the ice screw is clamped in a cylindrical fixture with an inside diameter slightly greater than the outside diameter of the ice piton. The top inside edge of the inside diameter of the clamping fixture shall have a radius of 1/8 ± 1/64 inch. The center of the eye shall be 2 ± 1/16 inches above the top of the clamping fixture. A pull shall be applied parallel to the shackle by means of a hook or link of 1/4-inch round steel bar, in a direction at a right angle to the tubing and shall be increased until the ice piton breaks or shows noticeable bending (permanent distortion) and recorded to the nearest 5 pounds. Any piton with a final pull strength of less than 565 pounds, shall be classified as a defect. The lot size shall be all pitons of one type offered for delivery at one time. The inspection level shall be S-2. Any test failure shall be cause for rejection of the lot. The test shall be performed at room temperature. Any industrial testing machine may be used to determine compliance with this requirement.

4. PACKAGING

4.1 Preservation. Pitons that are not furnished with a protective finish, shall be cleaned by process C-1, dried, and then coated with type P-1 preservative in accordance with MIL-P-116.

4.2 Packaging. Eight ice pitons shall be packaged in an intermediate box. The intermediate box shall be snug-fitting and fabricated from 200-pound test corrugated fiberboard.

4.3 Packing. One hundred twenty-eight ice pitons of one type, packaged as specified, shall be packed in a close-fitting corrugated fiberboard shipping container conforming to style RSC, type CF, class domestic, grade 275 of PPP-B-636. The shipping container shall be completely filled with no voids and of a size that can be palletized in accordance with MIL-STD-147. Each shipping container shall be closed in accordance with the appendix of PPP-B-636.

4.4 Palletization. When specified (see 5.2), pitons, packed as specified, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means C and D, or film bonding means F or G, in accordance with MIL-STD-147.

4.5 Marking. In addition to any special markings required by the contract or purchase order, intermediate boxes, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129.

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5. NOTES

5.1 Intended use. The pitons are intended to provide security for men or equipment in operations involving steep ascents by screwing the pitons into ice formations.

5.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Type required (see 1.1).
c. When palletization is required (see 4.4).

5.3 Sources of documents.

5.3.1 Source of government documents. Copies of military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D, 700 Robbins Avenue
Philadelphia, PA 19111-5094

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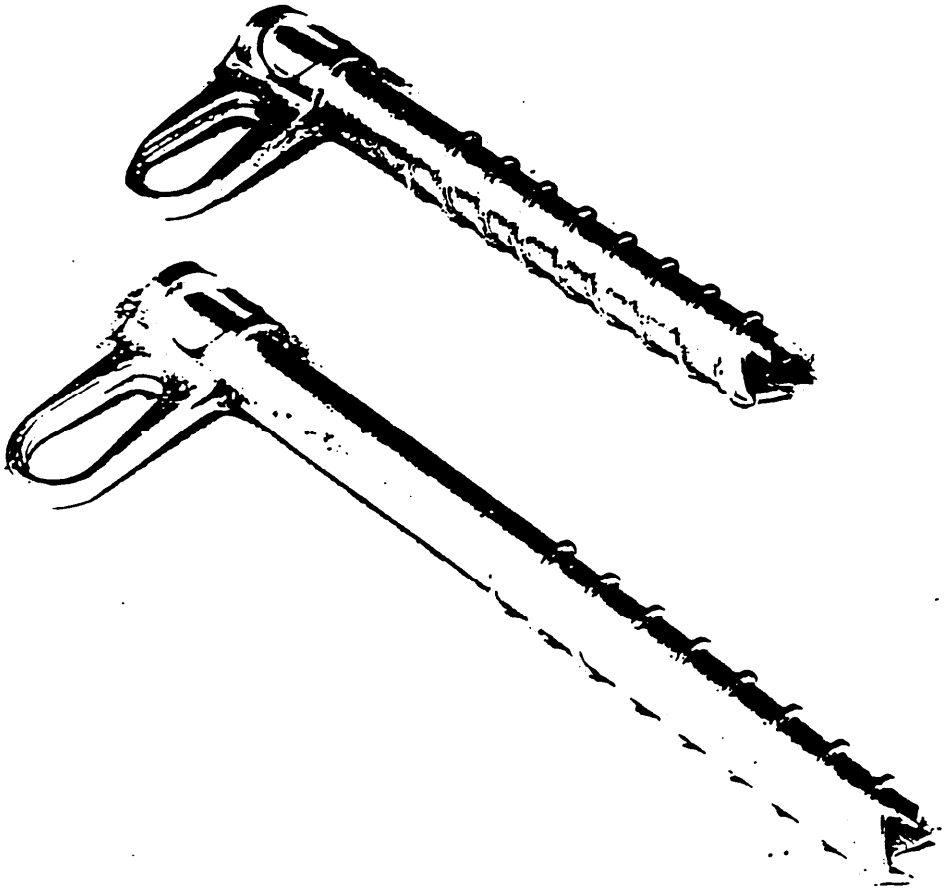


FIGURE 1. Piton, Ice.

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